

- It uses a *pull* technology, where the onus is on the client to pull configuration details from the server, instead of the server pushing to all clients.
- Completely encrypted communication and data transfer.
- It is available on all UNIX/GNU/Linux platforms:
 - Red Hat/Fedora and SuSE packages: *puppet* and *puppet-server*
 - Debian/Ubuntu flavours: *puppet* and *puppetmaster*
 The Puppet wiki's downloading page (see *References #2*) should give you more details.

Installing Puppet

There are three methods of installing Puppet (the version you should install is 2.6.x), and the preferred method varies by distribution:

- I. Source (un-tar and compile)
- II. Package managers or pre-built packages
- III. Ruby GEM

I would always prefer the second method that uses pre-built packages, since the others might need a lot of tweaking, depending on the OS. Even configuration files won't be in the right places, if not configured properly.

Pre-built packages

RPM/DEB files

If your distribution doesn't have a package manager with automatic dependency resolution, then download (see *References #2*) the following RPMs or DEBs, and install them in this order:

- a) Ruby (if not already installed, or not the latest version)
- b) *Facter*
- b) Puppet Master (for the server) or Puppet (client)

Yum

I would always go with this; it has the facility of even creating a repository for i386 or x64 architectures. This can be done in two ways:

- Fedora (*References #3*) and EPEL (*References #4*) Yum repositories already have it as part of the Fedora project. This can be downloaded and installed as RPMs, which will create the right repository, and you can install from there.
- I would prefer downloading the RPMs one by one, testing them for their dependencies, and creating my own Yum repository with the *createrepo* command. Please refer to online tutorials to set up Yum.

DEB packages for Ubuntu

Please don't install the default packages that come with Ubuntu—they are legacy packages. Download the source tarball and install from it as shown below:

```
$ wget http://puppetlabs.com/downloads/puppet/puppet-latest.tgz
$ untar and install it
$ gzip -d -c puppet-latest.tgz | tar xf -
```

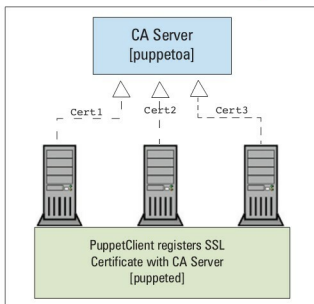


Figure 1: "Certificate Authorisation Request"

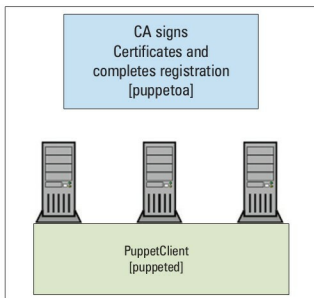


Figure 2: "Certificate Registration"

```
$ cd puppet-*
$ sudo ruby install.rb # or become root and run install.rb
```

Three major commands

- I) *puppetmasterd* (server daemon that runs as the Puppet user)
- II) *puppetca* (CA Server; part of the puppetmasterd)
- III) *puppetd* (client daemon)

Setting up PuppetMaster DNS resolution

Either the local */etc/hosts* entry has to have an alias 'puppet', or the global DNS server must have a CNAME that points to the Puppet server. Why? This architecture only understands